

2027 SEAK Pink Salmon Forecast

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Objective

To forecast the Southeast Alaska (SEAK) pink salmon commercial harvest in 2027. This document is for guidance as to what was done for the current forecast year. It is for internal use only.

Executive Summary

Forecasts were developed using an approach originally described in Wertheimer et al. (2006), and modified in Orsi et al. (2016) and Murphy et al. (2019), but assuming a log-normal error structure (Miller et al. 2022). This approach is based on a multiple linear regression model with either; 1) the raw juvenile pink salmon CPUE (`adj_raw_pink_log`) or 2) the vessel-adjusted juvenile pink salmon CPUE (i.e., adjusted by the pooled-species vessel calibration coefficient for the Cobb; vessel-adjusted CPUE) as a proxy for abundance. Additional variables included a vessel factor to account for the survey vessels through time, an odd and even year factor to account for potential odd and even year cycles of abundance, and temperature data from either the Southeast Alaska Coastal Monitoring Survey (SECM; Piston et al. 2021; `ISTI20_JJ`) or from satellite-derived sea surface temperature (SST) data (Huang et al. 2017).

There were 54 individual models considered:

- `adj_raw_pink_log` model, an odd/even year factor, and a vessel factor (`m1`);
- `adj_raw_pink_log` model, an odd/even year factor, a vessel factor, and temperature data from the the SECM survey (`m2`);
- `adj_raw_pink_log` models with an odd/even year factor, a vessel factor, and satellite SST data (`m3-m18`);
- `adj_raw_pink_log` model with a vessel interaction, an odd/even year factor, and a vessel factor (`m1i`);

- adj_raw_pink_log model with a vessel interaction, an odd/even year factor, a vessel factor, and temperature data from the the SECM survey (m2i);
- adj_raw_pink_log models with a vessel interaction, an odd/even year factor, a vessel factor, and satellite SST data (m3i-m18i);
- vessel-adjusted CPUE and an odd/even year factor (m1a);
- vessel-adjusted CPUE, an odd/even year factor, and temperature data from the SECM survey (m2a); and
- vessel-adjusted CPUE models with an odd/even year factor, and satellite-derived SST data (m3a-m18a).

The adj_raw_pink_log variable is defined as the natural log transformed juvenile pink salmon catches ($\ln \text{CPUE}+1$) by haul, standardized to a 20-minute trawl set. The CPUE values were averaged over all 8 stations by month; data from whichever month (June or July) had the highest log transformed CPUE is then used as the juvenile abundance index for that particular year. This CPUE variable is not vessel-calibrated. The Stellar and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the assessment models that incorporated an adj_raw_pink_log variable (Table 1).

The vessel-adjusted CPUE variable is defined as the natural log transformed juvenile pink salmon catches ($\ln \text{CPUE}+1$) by haul, standardized to a 20-minute trawl set. The CPUE was then multiplied by a pooled-species vessel calibration coefficient, based on calibration of the fishing power of various vessels used in the SECM project to the fishing power of the NOAA ship John N. Cobb, the vessel used for the initial 11 years of the project. The CPUE values were averaged over all 8 stations by month; data from whichever month (June or July) had the highest log transformed CPUE is then used as the juvenile abundance index for that particular year (Table 1). As there was not a SECM survey in May of 2025 or 2026, temperature data from May was unavailable. Therefore, the temperature variable from the SECM survey (ISTI20_JJ) was the average from June and July. In prior forecasting years, the ISTI variable was the average from May through July.

The model performance metric one-step ahead mean absolute percent error (MAPE) for the last five years (MAPE5; forecast years 2021 through 2025) was used to evaluate the forecast accuracy of the 54 individual models, the AICc values were calculated for each model to prevent over-parameterization of the model, and the adjusted R-squared values, significant terms, and overall p-value of the model were used to the determine fit.

Based upon the performance metric the 5-year MAPE, the AICc values, significant parameters in the models, and the adjusted R-squared values, model mxx (a model that included xxx; Appendix C) was the best performing model and the 2027-forecast using this model has a point estimate of xx3 million fish (80% prediction interval: xx to xx million fish).

Analysis

Individual, Multiple Linear Regression Models

Biophysical variables based on data from Southeast Alaska (Table 1) were used to forecast the harvest of adult pink salmon in Southeast Alaska, one year in advance, using individual, multiple linear regression models with two main structures (structure 1: models m1-m18, m1i-m18i; structure 2: models m1a-m18a). The response variable (Y ; Southeast Alaska adult pink salmon harvest in millions), the `adj_raw_pink_log` and the vessel-adjusted CPUE were log transformed in the model, but temperature data were not. The forecast ($\hat{Y}_i$), and 80% prediction intervals (based on output from program R; R Core Team 2025) from the 54 regression models were exponentiated and bias-corrected (Miller 1984),

$$\hat{F}_i = \exp(\hat{Y}_i + \frac{\sigma_i^2}{2}), \quad (1)$$

where $\hat{F}_i$ is the preseason forecast (for each model i) in millions of fish, and σ_i is the variance (for each model i).

Model structure #1 (m1i-m18i)

The first model structure, model m1i, consisted of the predictor variable juvenile `adj_raw_pink_log` (CPUE) with a vessel factor interaction, an odd/even year factor, and a vessel factor (model m1i),

$$E(Y) = \beta_0 + \beta_1 X_{V1} + \beta_2 X_{V2} + \beta_3 X_O + \beta_4 CPUE + \beta_5 (X_{V1} \times CPUE) + \beta_6 (X_{V2} \times CPUE) + \epsilon. \quad (2)$$

The reference category is an even year with the R/V Cobb. The vessel factor and even/odd year factor adjust the intercept, and the vessel interaction terms adjust the model slope. For example, during even years when the research vessel is the Cobb, the model simplifies to $E(Y) = \beta_0 + \beta_4 CPUE + \epsilon$. Although the simplest model does not contain a temperature variable, including temperature data with CPUE has been shown to result in a substantial improvement in the accuracy of model predictions (Murphy et al. 2019). The temperature index for models m2i-m18i was either the SECM survey Icy Strait temperature Index (ISTI20_JJ; Murphy et al. 2019) or one of the 16 satellite-derived SST data (Huang et al. 2017; definitions in Appendix A).

Model structure #1 (m1-m18)

Models m1 through m18 were a simplified version of models m1i through m18i and did not include the vessel interaction term,

$$E(Y) = \beta_0 + \beta_1 X_{V1} + \beta_2 X_{V2} + \beta_3 X_O + \beta_4 CPUE + \epsilon. \quad (3)$$

Models m1i through m18i estimate vessel-specific slopes for pink log abundance and additive effects of vessel and odd/even factor, while models m1 through m18 estimate a single common slope for pink log abundance, plus additive effects of vessel and odd/even factor, ignoring vessel-specific slope variability.

Model structure #2 (m1a-m18a)

The second model structure, model m1a, consisted of the predictor variable vessel-adjusted CPUE (CPUE), and an odd/even year factor,

$$E(Y) = \beta_0 + \beta_2 X_O + \beta_3 CPUE + \epsilon. \quad (4)$$

The reference category (i.e., reference intercept) is an even year. The temperature index for models m2a-m18a was either the SECM survey Icy Strait temperature Index (ISTI20_JJ; Murphy et al. 2019) or one of the 16 satellite-derived SST data (Huang et al. 2017).

Performance Metric: 5-Year One Step Ahead MAPE (MAPE5)

The model performance metric MAPE is a standard statistical metric used to measure the accuracy of forecasting and prediction models. It calculates the average of the absolute percentage differences between predicted values and actual outcomes. Because it measures error as a relative percentage rather than a raw number, a lower MAPE indicates higher accuracy, while a higher MAPE indicates a less reliable model. The MAPE value can be compared across the 54 models.

The model summary results using the performance metric 5-year one-step ahead MAPE are shown in Table 2, Table 3, and Table 4; the smallest value is the preferred model. The performance metric one-step ahead MAPE was calculated as follows.

1. Estimate the regression parameters at time $t-1$ from data up to time $t-1$.
2. Make a prediction of $\hat{Y}_t$ at time t based on the predictor variables at time t and the estimate of the regression parameters at time $t-1$ (i.e., the fitted regression equation).
3. Calculate the MAPE based on the prediction of $\hat{Y}_t$ at time t and the observed value of Y_t at time t ,

$$\text{MAPE} = \left| \frac{\exp(Y_t) - \exp(\hat{Y}_t + \frac{\sigma_t^2}{2})}{\exp(Y_t)} \right|. \quad (5)$$

4. For each individual model, average the MAPEs calculated from the forecasts,

$$\frac{1}{n} \sum_{t=1}^n \left| \frac{\exp(Y_t) - \exp(\hat{Y}_t + \frac{\sigma_t^2}{2})}{\exp(Y_t)} \right|, \quad (4)$$

where n is the number of forecasts in the average (5 forecasts for the 5-year MAPE and 10 forecasts for the 10-year MAPE). For example, to calculate the five year one-step-ahead MAPE for model m1 for the 2022 forecast, use data up through year 2016 (e.g., data up through year 2016 is $t - 1$ and the forecast is for t , or year 2017). Then, calculate a MAPE based on the 2017 forecast and the observed pink salmon harvest in 2017 using equation 3. Next, use data up through year 2017 (e.g., data up through year 2017 is $t - 1$ and the forecast is for year 2018; t) and calculate a MAPE based on the 2018 forecast and the observed pink salmon harvest in 2018 using equation 3. Repeat this process for each subsequent year through year 2020 to forecast 2021. Finally, average the five MAPEs to calculate a five year one-step-ahead MAPE for model m1. As the results of the 5-year MAPEs with or without the forecast bias adjustment have been similar (i.e., the model performance order did not change whether the five year one-step-ahead MAPE or the bias-corrected five year one-step-ahead MAPE was compared), for simplicity, the bias adjustment for the forecast was not used in the calculation of the five year one-step-ahead MAPE for model comparison.

Akaike Information Criterion Corrected for Small Sample Sizes (AICc)

Hierarchical models were compared with the AICc criterion. The best fit models, according to the AICc criterion, is one that explains the greatest amount of variation with the fewest independent variables (i.e., the most parsimonious; Table 2, Table 3, table 4). Lower AICc values are preferred, and the AICc criterion penalizes models that use more parameters. Comparing the AICc values of two hierarchical models, a $\Delta_i \leq 2$ suggests that the two models are essentially the same, and the most parsimonious model should be chosen (Burnham and Anderson 2004). If the $\Delta_i > 2$, the model with the lower AICc value should be chosen. As the data set of the response variable is reduced for the model data set for models m1-m18 and models mli-m18i compared to the model set m1a-m18a, AICc can only be used to compare within the nested models (e.g., among m1-m18, mli-m18i).

Adjusted R-squared

The adjusted R-squared is a modified version of the R-squared value that adjusts for the number of predictors in the model and prevents overfitting by balancing model performance and

complexity. A higher adjusted R-squared value is preferred. As the adjusted R-squared metric depends on both sample size and the variance of the dependent variable in the data set, the adjusted R-squared can only be compared for the models with the same response variable data set. As the data set for the models m1-m18 and m1i-m18i is reduced, the adjusted R-squared can only be compared within the model group (i.e., among m1-m18, m1i-m18i) and not with models m1a through m18a.

Results

Models m1-m18

Models m1i-m18i

Models m2a-m18a

Based upon the 5-year MAPE performance metric, the AICc values, significant parameters in the models, and the adjusted R-squared values, model mxx (a model that included xxxx Table 5; Appendix C) was the best performing model. The 2027-forecast using this model has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).

Explain that the interaction term was not significant by AICc, anova, R squared for m5 and m5i.....

Table 1: Summary of the data used for the 2027-forecast. The Stellar and Chellissa vessels were only used for one year each, 2008 and 2009, respectively, and so those two years are not used in the assessment models that incorporated an `adj_raw_pink_log` variable. The CPUE variable is the vessel-adjusted CPUE, JYear is the juvenile year, and Year is the calendar year.

JYear	Year	Harvest	odd_even_factor	vessel	adj_raw_pink_log	CPUE
1997	1998	42.4	even	Cobb	2.48	2.48
1998	1999	77.8	odd	Cobb	5.62	5.62
1999	2000	20.2	even	Cobb	1.60	1.60
2000	2001	67.0	odd	Cobb	3.73	3.73
2001	2002	45.3	even	Cobb	2.87	2.87
2002	2003	52.5	odd	Cobb	2.78	2.78
2003	2004	45.3	even	Cobb	3.08	3.08
2004	2005	59.1	odd	Cobb	3.90	3.90
2005	2006	11.6	even	Cobb	2.04	2.04
2006	2007	44.8	odd	Cobb	2.57	2.57
2007	2008	15.9	even	Cobb	1.17	1.17

JYear	Year	Harvest	odd_even_factor	vessel	adj_raw_pink_log	CPUE
2008	2009	38.0	odd	Steller	2.45	2.32
2009	2010	24.1	even	Chellissa	3.16	2.33
2010	2011	58.9	odd	NW Explorer	5.57	4.11
2011	2012	21.3	even	NW Explorer	1.97	1.45
2012	2013	94.7	odd	NW Explorer	4.78	3.52
2013	2014	37.2	even	NW Explorer	2.91	2.14
2014	2015	35.1	odd	NW Explorer	5.18	3.82
2015	2016	18.4	even	NW Explorer	3.33	2.45
2016	2017	34.7	odd	NW Explorer	5.90	4.35
2017	2018	8.1	even	NW Explorer	0.47	0.35
2018	2019	21.1	odd	Medeia	1.39	1.17
2019	2020	8.1	even	Medeia	1.36	1.14
2020	2021	48.5	odd	Medeia	2.56	2.15
2021	2022	18.3	even	Medeia	1.04	0.87
2022	2023	48.5	odd	Medeia	1.72	1.45
2023	2024	20.1	even	Medeia	1.42	1.19
2024	2025	21.4	odd	Medeia	1.98	1.66
2025	2026	20.4	even	Medeia	1.53	1.29
2026	2027	NA	odd	Medeia	4.18	3.51

Table 2: Summary of the performance metrics for the 18 regression models with adj_raw_pink_log and a vessel-specific slope. The AICc value is the change in AICc value.

Terms	Model	Fit	Fit_LPI	Fit_UPI	AdjR2	MAPE5	AICc_change
Chatham_SST_May	m3i	401.2	146.9	1095.6	0.81	32.9	0.3
Icy_Strait_SST_May	m7i	439.8	154.4	1253.3	0.80	34.4	2.0
IST120_JJ	m2i	207.9	62.1	696.5	0.70	34.9	12.3
Chatham_SST_AMJ	m5i	237.2	84.7	664.6	0.78	35.0	3.7
SEAK_SST_May	m15i	314.4	113.2	873.4	0.79	36.3	2.2
NSEAK_SST_May	m11i	386.2	142.8	1044.3	0.81	36.6	0.0
SEAK_SST_AMJ	m17i	192.3	67.4	548.7	0.77	38.1	5.2
NSEAK_SST_AMJ	m13i	249.8	89.5	697.2	0.79	38.9	3.3
no temperature index included	m1i	112.5	27.4	463.0	0.57	39.9	19.1
Icy_Strait_SST_AMJ	m9i	287.3	97.8	843.4	0.77	40.6	5.2
Chatham_SST_AMJJ	m6i	176.2	56.5	549.8	0.73	41.8	9.8
SEAK_SST_AMJJ	m18i	148.3	47.6	461.4	0.73	41.9	10.0
NSEAK_SST_AMJJ	m14i	190.4	61.8	586.4	0.74	45.0	8.9
SEAK_SST_MJJ	m16i	129.8	38.1	442.7	0.68	46.5	14.4
Icy_Strait_SST_AMJJ	m10i	215.8	66.5	700.3	0.72	46.7	10.8
Chatham_SST_MJJ	m4i	159.3	45.6	556.4	0.67	47.1	15.0
NSEAK_SST_MJJ	m12i	164.7	48.5	558.6	0.69	50.7	13.6
Icy_Strait_SST_MJJ	m8i	184.3	52.1	652.6	0.67	51.4	15.0

Table 3: Summary of the performance metrics for the 18 regression models with adj_raw_pink_log. The AICc value is the change in AICc value.

Terms	Model	Fit	Fit_LPI	Fit_UPI	AdjR2	MAPE5	AICc_change
Icy_Strait_SST_May	m7	111.7	66.3	188.1	0.77	20.6	0.7
Chatham_SST_May	m3	101.9	61.6	168.4	0.78	22.3	0.0
Chatham_SST_AMJ	m5	90.6	55.1	149.1	0.77	25.2	0.7
SEAK_SST_May	m15	83.3	50.4	137.5	0.76	25.5	1.9
NSEAK_SST_May	m11	95.8	58.2	157.6	0.78	26.2	0.1
SEAK_SST_AMJ	m17	77.2	46.7	127.6	0.76	28.0	2.6
NSEAK_SST_AMJ	m13	91.6	55.5	151.3	0.77	28.3	1.0
Icy_Strait_SST_AMJ	m9	107.4	63.5	181.8	0.77	30.0	1.6
no temperature index included	m1	56.5	29.9	106.7	0.59	30.0	14.3
Chatham_SST_AMJJ	m6	92.7	53.5	160.6	0.73	33.3	5.2
ISTI20_JJ	m2	90.6	51.1	160.5	0.71	33.3	7.2
SEAK_SST_AMJJ	m18	82.1	47.8	141.0	0.73	33.4	5.7
NSEAK_SST_AMJJ	m14	95.6	55.2	165.8	0.73	35.7	4.9
Icy_Strait_SST_AMJJ	m10	101.5	57.1	180.4	0.72	37.0	6.2
SEAK_SST_MJJ	m16	76.6	43.1	136.1	0.69	37.7	9.3
Chatham_SST_MJJ	m4	86.2	47.4	156.8	0.68	37.8	9.7
NSEAK_SST_MJJ	m12	87.0	48.4	156.5	0.69	40.7	8.8
Icy_Strait_SST_MJJ	m8	88.6	48.4	162.2	0.68	40.9	9.8

Table 4: Summary of the performance metrics for the 18 regression models with vessel-adjusted CPUE. The AICc value is the change in AICc value.

Terms	Model	Fit	Fit_LPI	Fit_UPI	AdjR2	MAPE5	AICc_change
Chatham_SST_AMJ	m5a	69.5	45.6	106.1	0.77	20.5	0.8
ISTI20_JJ	m2a	67.5	41.1	110.8	0.69	21.2	9.8
Icy_Strait_SST_AMJ	m9a	80.4	51.9	124.8	0.77	21.2	1.6
NSEAK_SST_AMJ	m13a	66.7	43.5	102.3	0.77	22.4	1.6
Chatham_SST_May	m3a	74.7	49.0	113.7	0.78	23.3	0.0
NSEAK_SST_May	m11a	69.2	45.5	105.3	0.77	23.6	0.5
Icy_Strait_SST_May	m7a	80.6	52.3	124.3	0.77	24.1	0.9
SEAK_SST_May	m15a	60.6	39.4	93.1	0.76	24.6	2.4
SEAK_SST_AMJ	m17a	57.2	37.0	88.5	0.75	24.8	3.4
no temperature index included	m1a	56.8	33.2	97.3	0.62	25.3	13.9
Icy_Strait_SST_AMJJ	m10a	79.5	49.4	128.1	0.73	25.5	6.0
Chatham_SST_AMJJ	m6a	70.1	44.2	111.1	0.73	25.8	5.6
NSEAK_SST_AMJJ	m14a	67.7	42.6	107.5	0.73	27.1	6.1
SEAK_SST_AMJJ	m18a	59.0	37.1	93.8	0.72	29.1	6.9
Icy_Strait_SST_MJJ	m8a	75.4	45.7	124.3	0.70	29.8	9.0
Chatham_SST_MJJ	m4a	69.3	42.4	113.4	0.70	30.0	9.2
NSEAK_SST_MJJ	m12a	65.5	40.4	106.4	0.70	30.7	8.8
SEAK_SST_MJJ	m16a	58.1	35.9	94.1	0.70	31.4	9.3

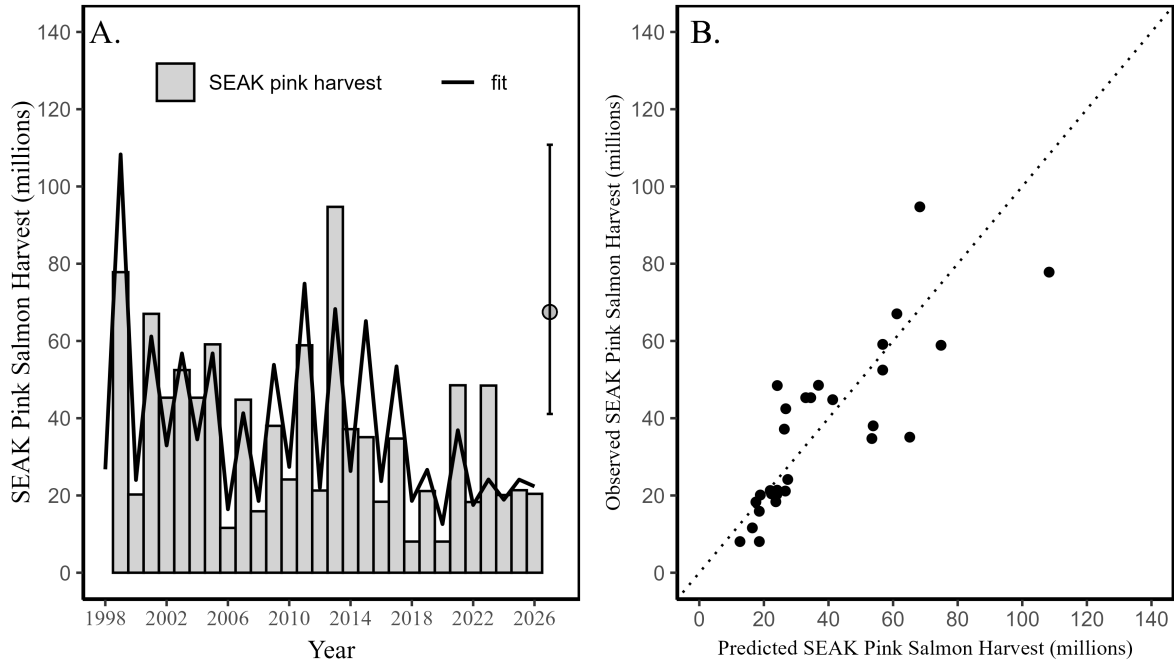


Figure 1: A. SEAK pink salmon harvest (millions) by year with the model fit (line) based upon the best performing model (model m2a). The predicted 2027 forecast (67.5 million fish) is symbolized as a grey circle with an 80% prediction interval (41.1 to 110.8 million fish). B. SEAK pink salmon harvest (millions) against the fitted values from model m2a by year. The dotted line is a one to one reference line.

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Appendix A

Variable definitions

adj_raw_pink_log: First, the raw juvenile pink salmon catch by year, month, haul, and area was adjusted to a 20 minute haul (raw_pink_catch). Then, these values were log-transformed (i.e., $\ln(\text{raw_pink_catch} + 1)$). Next, the maximum value in either June or July, whichever month had the highest in a given year, was used as the adj_raw_pink_log value for the year (i.e., $\text{adj_raw_pink_log} = \text{maximum}(\ln(\text{raw_pink_catch} + 1))$ in either June or July).

vessel-adjusted CPUE: The average $\ln(\text{CPUE}+1)$ for juvenile pink salmon catches in either June or July, whichever month had the highest average in a given year, where effort was a standard trawl haul (20 minutes). The juvenile CPUE data was adjusted using vessel calibration factors to account for differences in fishing power among vessels.

ISTI20_JJ: The average 20-m integrated water column temperature at the eight stations in Icy Strait (Icy Strait and Upper Chatham transects) sampled during the SECM surveys in June and July of each year (in degrees Celsius). In 2025, there was no survey in May. Therefore, this temperature index no longer includes May in the time series temperature index.

Satellite SST variables

Icy_Strait_SST_May: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in May.

Icy_Strait_SST_MJJ: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in May through July.

Icy_Strait_SST_AMJ: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in April through June.

Icy_Strait_SST_AMJJ: The Icy Strait region encompasses waters of Icy Strait from the east end of Lemesurier Island to a line from Point Couverden south to Point Augusta. This variable is the average SST in April through July.

Chatham_SST_May: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, and south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in May.

Chatham_SST_MJJ: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in May through July.

Chatham_SST_AMJ: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in April through June. The last time this variable was incorporated in the forecasting process was the 2023 forecast.

Chatham_SST_AMJJ: The Chatham and Icy Straits region encompasses waters of Chatham and Icy Straits east of Lemesurier Island to Point Couverden, south to the approximate latitude of 56.025 degrees north (roughly Cape Decision off Kuiu Island). This variable is the average SST in April through July.

NSEAK_SST_May: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in May.

NSEAK_SST_MJJ: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in May through July.

NSEAK_SST_AMJ: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in April through June.

NSEAK_SST_AMJJ: The NSEAK region encompasses northern Southeast Alaska from 59.475 to 56.075 degrees north latitude (approximately Districts 9 through 15, and District 13 inside area only; northern Southeast Inside subregion for Southeast Alaska (NSEI)). This variable is the average SST in April through July.

SEAK_SST_May: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in May. The last time this variable was incorporated in the forecasting process was the 2023 forecast.

SEAK_SST_MJJ: The SEAK region encompasses northern Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in May through July.

SEAK_SST_AMJ: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in April through June.

SEAK_SST_AMJJ: The SEAK region encompasses Southeast Alaska from 59.475 to 54.725 degrees north latitude. This variable is the average SST in April through July.

Appendix B

Table 5: Parameter estimates for the 18 individual models for Models m1i-m18i.

Model	Term	Estimate	Standard Error	Statistic	p value
m1i	(Intercept)	2.574	0.362	7.114	0.000
m1i	as.factor(vessel)Medeia	-0.493	0.694	-0.711	0.485
m1i	as.factor(vessel)NW Explorer	-0.144	0.501	-0.288	0.776
m1i	adj_raw_pink_log	0.311	0.132	2.362	0.028
m1i	as.factor(odd_even_factor)odd	0.358	0.245	1.463	0.159
m1i	as.factor(vessel)Medeia:adj_raw_pink_log	0.213	0.387	0.551	0.588
m1i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.094	0.142	-0.664	0.515
m2i	(Intercept)	7.633	1.634	4.672	0.000
m2i	as.factor(vessel)Medeia	-0.518	0.577	-0.898	0.381
m2i	as.factor(vessel)NW Explorer	-0.482	0.430	-1.120	0.277
m2i	adj_raw_pink_log	0.454	0.119	3.830	0.001
m2i	ISTI20_JJ	-0.526	0.167	-3.151	0.005
m2i	as.factor(odd_even_factor)odd	-0.010	0.235	-0.042	0.967
m2i	as.factor(vessel)Medeia:adj_raw_pink_log	0.294	0.323	0.911	0.374
m2i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.056	0.118	-0.476	0.640
m3i	(Intercept)	6.305	0.770	8.193	0.000
m3i	as.factor(vessel)Medeia	-0.536	0.462	-1.160	0.261
m3i	as.factor(vessel)NW Explorer	0.004	0.335	0.012	0.991
m3i	adj_raw_pink_log	0.465	0.093	5.009	0.000
m3i	Chatham_SST_May	-0.552	0.108	-5.105	0.000
m3i	as.factor(odd_even_factor)odd	0.113	0.170	0.667	0.513
m3i	as.factor(vessel)Medeia:adj_raw_pink_log	0.467	0.262	1.778	0.091
m3i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.096	0.094	-1.018	0.321
m4i	(Intercept)	6.512	1.499	4.345	0.000
m4i	as.factor(vessel)Medeia	-0.246	0.613	-0.401	0.693
m4i	as.factor(vessel)NW Explorer	-0.096	0.438	-0.219	0.829
m4i	adj_raw_pink_log	0.386	0.118	3.261	0.004
m4i	Chatham_SST_MJJ	-0.422	0.157	-2.688	0.015
m4i	as.factor(odd_even_factor)odd	0.255	0.217	1.173	0.255
m4i	as.factor(vessel)Medeia:adj_raw_pink_log	0.180	0.338	0.531	0.601
m4i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.101	0.124	-0.816	0.424
m5i	(Intercept)	6.911	0.985	7.019	0.000
m5i	as.factor(vessel)Medeia	-0.357	0.493	-0.726	0.477
m5i	as.factor(vessel)NW Explorer	0.001	0.356	0.003	0.998
m5i	adj_raw_pink_log	0.440	0.098	4.515	0.000
m5i	Chatham_SST_AMJ	-0.612	0.134	-4.563	0.000

Model	Term	Estimate	Standard Error	Statistic	p value
m5i	as.factor(odd_even_factor)odd	0.149	0.180	0.832	0.416
m5i	as.factor(vessel)Medeia:adj_raw_pink_log	0.314	0.275	1.141	0.268
m5i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.101	0.100	-1.007	0.327
m6i	(Intercept)	7.107	1.299	5.469	0.000
m6i	as.factor(vessel)Medeia	-0.180	0.557	-0.323	0.750
m6i	as.factor(vessel)NW Explorer	-0.054	0.398	-0.136	0.893
m6i	adj_raw_pink_log	0.422	0.109	3.876	0.001
m6i	Chatham_SST_AMJJ	-0.553	0.155	-3.577	0.002
m6i	as.factor(odd_even_factor)odd	0.201	0.199	1.007	0.326
m6i	as.factor(vessel)Medeia:adj_raw_pink_log	0.183	0.307	0.595	0.559
m6i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.112	0.112	-0.993	0.333
m7i	(Intercept)	5.975	0.745	8.017	0.000
m7i	as.factor(vessel)Medeia	-0.606	0.477	-1.270	0.219
m7i	as.factor(vessel)NW Explorer	-0.099	0.344	-0.287	0.777
m7i	adj_raw_pink_log	0.448	0.095	4.729	0.000
m7i	Icy_Strait_SST_May	-0.531	0.110	-4.841	0.000
m7i	as.factor(odd_even_factor)odd	0.167	0.173	0.965	0.346
m7i	as.factor(vessel)Medeia:adj_raw_pink_log	0.487	0.271	1.793	0.089
m7i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.062	0.097	-0.637	0.532
m8i	(Intercept)	6.045	1.333	4.536	0.000
m8i	as.factor(vessel)Medeia	-0.420	0.607	-0.692	0.497
m8i	as.factor(vessel)NW Explorer	-0.162	0.438	-0.371	0.715
m8i	adj_raw_pink_log	0.367	0.117	3.136	0.005
m8i	Icy_Strait_SST_MJJ	-0.361	0.135	-2.681	0.015
m8i	as.factor(odd_even_factor)odd	0.291	0.216	1.351	0.192
m8i	as.factor(vessel)Medeia:adj_raw_pink_log	0.237	0.338	0.702	0.491
m8i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.084	0.124	-0.681	0.504
m9i	(Intercept)	6.609	0.973	6.794	0.000
m9i	as.factor(vessel)Medeia	-0.441	0.506	-0.872	0.394
m9i	as.factor(vessel)NW Explorer	-0.105	0.365	-0.288	0.777
m9i	adj_raw_pink_log	0.418	0.099	4.216	0.000
m9i	Icy_Strait_SST_AMJ	-0.586	0.136	-4.310	0.000
m9i	as.factor(odd_even_factor)odd	0.228	0.181	1.256	0.224
m9i	as.factor(vessel)Medeia:adj_raw_pink_log	0.343	0.284	1.208	0.242
m9i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.068	0.103	-0.662	0.516
m10i	(Intercept)	6.523	1.198	5.444	0.000
m10i	as.factor(vessel)Medeia	-0.344	0.563	-0.610	0.549
m10i	as.factor(vessel)NW Explorer	-0.127	0.405	-0.314	0.757
m10i	adj_raw_pink_log	0.403	0.110	3.666	0.002
m10i	Icy_Strait_SST_AMJJ	-0.480	0.141	-3.399	0.003
m10i	as.factor(odd_even_factor)odd	0.254	0.201	1.267	0.221

Model	Term	Estimate	Standard Error	Statistic	p value
m10i	as.factor(vessel)Medeia:adj_raw_pink_log	0.238	0.313	0.760	0.457
m10i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.095	0.115	-0.830	0.417
m11i	(Intercept)	6.018	0.709	8.486	0.000
m11i	as.factor(vessel)Medeia	-0.479	0.459	-1.044	0.310
m11i	as.factor(vessel)NW Explorer	0.072	0.334	0.215	0.832
m11i	adj_raw_pink_log	0.446	0.091	4.901	0.000
m11i	NSEAK_SST_May	-0.516	0.100	-5.161	0.000
m11i	as.factor(odd_even_factor)odd	0.156	0.167	0.938	0.360
m11i	as.factor(vessel)Medeia:adj_raw_pink_log	0.469	0.261	1.800	0.088
m11i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.107	0.094	-1.136	0.270
m12i	(Intercept)	6.514	1.383	4.710	0.000
m12i	as.factor(vessel)Medeia	-0.132	0.604	-0.219	0.829
m12i	as.factor(vessel)NW Explorer	-0.011	0.429	-0.027	0.979
m12i	adj_raw_pink_log	0.390	0.115	3.379	0.003
m12i	NSEAK_SST_MJJ	-0.425	0.145	-2.923	0.009
m12i	as.factor(odd_even_factor)odd	0.264	0.211	1.252	0.226
m12i	as.factor(vessel)Medeia:adj_raw_pink_log	0.178	0.330	0.539	0.596
m12i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.115	0.121	-0.950	0.354
m13i	(Intercept)	6.733	0.935	7.197	0.000
m13i	as.factor(vessel)Medeia	-0.259	0.491	-0.528	0.604
m13i	as.factor(vessel)NW Explorer	0.096	0.356	0.268	0.791
m13i	adj_raw_pink_log	0.435	0.096	4.504	0.000
m13i	NSEAK_SST_AMJ	-0.602	0.130	-4.621	0.000
m13i	as.factor(odd_even_factor)odd	0.184	0.177	1.042	0.310
m13i	as.factor(vessel)Medeia:adj_raw_pink_log	0.317	0.273	1.162	0.260
m13i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.117	0.100	-1.168	0.257
m14i	(Intercept)	7.003	1.226	5.711	0.000
m14i	as.factor(vessel)Medeia	-0.084	0.553	-0.151	0.881
m14i	as.factor(vessel)NW Explorer	0.027	0.394	0.069	0.946
m14i	adj_raw_pink_log	0.422	0.107	3.943	0.001
m14i	NSEAK_SST_AMJJ	-0.548	0.148	-3.712	0.001
m14i	as.factor(odd_even_factor)odd	0.224	0.195	1.152	0.263
m14i	as.factor(vessel)Medeia:adj_raw_pink_log	0.188	0.302	0.623	0.541
m14i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.127	0.111	-1.142	0.268
m15i	(Intercept)	6.120	0.780	7.848	0.000
m15i	as.factor(vessel)Medeia	-0.388	0.479	-0.809	0.428
m15i	as.factor(vessel)NW Explorer	0.115	0.350	0.330	0.745
m15i	adj_raw_pink_log	0.460	0.096	4.791	0.000
m15i	SEAK_SST_May	-0.495	0.103	-4.800	0.000
m15i	as.factor(odd_even_factor)odd	0.145	0.175	0.830	0.417
m15i	as.factor(vessel)Medeia:adj_raw_pink_log	0.439	0.271	1.619	0.122

Model	Term	Estimate	Standard Error	Statistic	p value
m15i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.121	0.098	-1.238	0.231
m16i	(Intercept)	6.423	1.414	4.541	0.000
m16i	as.factor(vessel)Medeia	-0.015	0.623	-0.023	0.982
m16i	as.factor(vessel)NW Explorer	0.019	0.437	0.043	0.966
m16i	adj_raw_pink_log	0.389	0.117	3.318	0.004
m16i	SEAK_SST_MJJ	-0.397	0.142	-2.791	0.012
m16i	as.factor(odd_even_factor)odd	0.260	0.215	1.212	0.240
m16i	as.factor(vessel)Medeia:adj_raw_pink_log	0.136	0.335	0.404	0.690
m16i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.114	0.123	-0.933	0.362
m17i	(Intercept)	6.745	1.002	6.733	0.000
m17i	as.factor(vessel)Medeia	-0.150	0.512	-0.293	0.773
m17i	as.factor(vessel)NW Explorer	0.132	0.371	0.357	0.725
m17i	adj_raw_pink_log	0.445	0.101	4.412	0.000
m17i	SEAK_SST_AMJ	-0.564	0.131	-4.316	0.000
m17i	as.factor(odd_even_factor)odd	0.180	0.183	0.981	0.339
m17i	as.factor(vessel)Medeia:adj_raw_pink_log	0.275	0.282	0.972	0.343
m17i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.132	0.104	-1.270	0.220
m18i	(Intercept)	6.971	1.277	5.457	0.000
m18i	as.factor(vessel)Medeia	0.034	0.573	0.060	0.953
m18i	as.factor(vessel)NW Explorer	0.070	0.404	0.172	0.865
m18i	adj_raw_pink_log	0.422	0.110	3.852	0.001
m18i	SEAK_SST_AMJJ	-0.515	0.146	-3.533	0.002
m18i	as.factor(odd_even_factor)odd	0.225	0.199	1.132	0.272
m18i	as.factor(vessel)Medeia:adj_raw_pink_log	0.150	0.309	0.484	0.634
m18i	as.factor(vessel)NW Explorer:adj_raw_pink_log	-0.129	0.113	-1.141	0.268

Table 6: Parameter estimates for the 18 individual models for Models m1-m18.

Model	Term	Estimate	Standard Error	Statistic	p value
m1	(Intercept)	2.740	0.241	11.379	0.000
m1	as.factor(odd_even_factor)odd	0.423	0.226	1.870	0.075
m1	as.factor(vessel)Medeia	-0.236	0.232	-1.019	0.319
m1	as.factor(vessel)NW Explorer	-0.442	0.209	-2.109	0.047
m1	adj_raw_pink_log	0.243	0.090	2.697	0.013
m2	(Intercept)	7.707	1.559	4.943	0.000
m2	ISTI20_JJ	-0.524	0.163	-3.212	0.004
m2	as.factor(odd_even_factor)odd	0.065	0.220	0.295	0.771
m2	as.factor(vessel)Medeia	-0.100	0.199	-0.503	0.620
m2	as.factor(vessel)NW Explorer	-0.658	0.188	-3.498	0.002
m2	adj_raw_pink_log	0.409	0.092	4.468	0.000
m3	(Intercept)	6.174	0.789	7.830	0.000
m3	Chatham_SST_May	-0.509	0.114	-4.468	0.000
m3	as.factor(odd_even_factor)odd	0.246	0.170	1.444	0.163
m3	as.factor(vessel)Medeia	0.094	0.185	0.509	0.616
m3	as.factor(vessel)NW Explorer	-0.310	0.156	-1.980	0.061
m3	adj_raw_pink_log	0.379	0.073	5.206	0.000
m4	(Intercept)	6.703	1.456	4.603	0.000
m4	Chatham_SST_MJJ	-0.423	0.154	-2.750	0.012
m4	as.factor(odd_even_factor)odd	0.314	0.202	1.551	0.136
m4	as.factor(vessel)Medeia	-0.047	0.214	-0.219	0.829
m4	as.factor(vessel)NW Explorer	-0.416	0.184	-2.262	0.034
m4	adj_raw_pink_log	0.314	0.083	3.774	0.001
m5	(Intercept)	6.960	0.987	7.054	0.000
m5	Chatham_SST_AMJ	-0.594	0.137	-4.349	0.000
m5	as.factor(odd_even_factor)odd	0.242	0.173	1.399	0.177
m5	as.factor(vessel)Medeia	0.045	0.184	0.247	0.807
m5	as.factor(vessel)NW Explorer	-0.322	0.158	-2.040	0.054
m5	adj_raw_pink_log	0.362	0.072	5.006	0.000
m6	(Intercept)	7.271	1.280	5.680	0.000
m6	Chatham_SST_AMJJ	-0.549	0.153	-3.581	0.002
m6	as.factor(odd_even_factor)odd	0.264	0.188	1.407	0.174
m6	as.factor(vessel)Medeia	0.012	0.199	0.058	0.954
m6	as.factor(vessel)NW Explorer	-0.409	0.169	-2.417	0.025
m6	adj_raw_pink_log	0.343	0.078	4.401	0.000
m7	(Intercept)	5.832	0.732	7.965	0.000
m7	Icy_Strait_SST_May	-0.492	0.113	-4.354	0.000
m7	as.factor(odd_even_factor)odd	0.291	0.170	1.710	0.102
m7	as.factor(vessel)Medeia	0.089	0.187	0.474	0.641

Model	Term	Estimate	Standard Error	Statistic	p value
m7	as.factor(vessel)NW Explorer	-0.303	0.159	-1.912	0.070
m7	adj_raw_pink_log	0.385	0.074	5.171	0.000
m8	(Intercept)	6.195	1.280	4.839	0.000
m8	Icy_Strait_SST_MJJ	-0.361	0.132	-2.736	0.012
m8	as.factor(odd_even_factor)odd	0.359	0.200	1.793	0.087
m8	as.factor(vessel)Medeia	-0.116	0.208	-0.557	0.584
m8	as.factor(vessel)NW Explorer	-0.429	0.184	-2.330	0.030
m8	adj_raw_pink_log	0.305	0.082	3.700	0.001
m9	(Intercept)	6.636	0.944	7.028	0.000
m9	Icy_Strait_SST_AMJ	-0.573	0.136	-4.204	0.000
m9	as.factor(odd_even_factor)odd	0.316	0.172	1.833	0.081
m9	as.factor(vessel)Medeia	0.038	0.186	0.204	0.840
m9	as.factor(vessel)NW Explorer	-0.323	0.161	-2.014	0.057
m9	adj_raw_pink_log	0.363	0.074	4.917	0.000
m10	(Intercept)	6.661	1.168	5.702	0.000
m10	Icy_Strait_SST_AMJJ	-0.476	0.140	-3.406	0.003
m10	as.factor(odd_even_factor)odd	0.325	0.188	1.727	0.099
m10	as.factor(vessel)Medeia	-0.049	0.198	-0.248	0.806
m10	as.factor(vessel)NW Explorer	-0.428	0.172	-2.488	0.021
m10	adj_raw_pink_log	0.333	0.079	4.235	0.000
m11	(Intercept)	5.901	0.733	8.056	0.000
m11	NSEAK_SST_May	-0.472	0.106	-4.447	0.000
m11	as.factor(odd_even_factor)odd	0.289	0.169	1.715	0.101
m11	as.factor(vessel)Medeia	0.139	0.190	0.733	0.472
m11	as.factor(vessel)NW Explorer	-0.279	0.158	-1.764	0.092
m11	adj_raw_pink_log	0.354	0.071	5.001	0.000
m12	(Intercept)	6.669	1.358	4.910	0.000
m12	NSEAK_SST_MJJ	-0.419	0.143	-2.927	0.008
m12	as.factor(odd_even_factor)odd	0.327	0.198	1.655	0.113
m12	as.factor(vessel)Medeia	0.047	0.222	0.213	0.833
m12	as.factor(vessel)NW Explorer	-0.377	0.182	-2.071	0.051
m12	adj_raw_pink_log	0.308	0.081	3.809	0.001
m13	(Intercept)	6.761	0.951	7.111	0.000
m13	NSEAK_SST_AMJ	-0.576	0.134	-4.307	0.000
m13	as.factor(odd_even_factor)odd	0.281	0.172	1.637	0.116
m13	as.factor(vessel)Medeia	0.129	0.192	0.669	0.511
m13	as.factor(vessel)NW Explorer	-0.281	0.161	-1.746	0.095
m13	adj_raw_pink_log	0.345	0.071	4.837	0.000
m14	(Intercept)	7.126	1.224	5.821	0.000
m14	NSEAK_SST_AMJJ	-0.535	0.148	-3.628	0.002
m14	as.factor(odd_even_factor)odd	0.294	0.185	1.590	0.127

Model	Term	Estimate	Standard Error	Statistic	p value
m14	as.factor(vessel)Medeia	0.098	0.207	0.473	0.641
m14	as.factor(vessel)NW Explorer	-0.377	0.169	-2.228	0.037
m14	adj_raw_pink_log	0.331	0.076	4.338	0.000
m15	(Intercept)	6.006	0.807	7.439	0.000
m15	SEAK_SST_May	-0.451	0.109	-4.153	0.000
m15	as.factor(odd_even_factor)odd	0.277	0.175	1.581	0.129
m15	as.factor(vessel)Medeia	0.165	0.200	0.824	0.419
m15	as.factor(vessel)NW Explorer	-0.283	0.163	-1.729	0.098
m15	adj_raw_pink_log	0.358	0.074	4.850	0.000
m16	(Intercept)	6.615	1.379	4.796	0.000
m16	SEAK_SST_MJJ	-0.395	0.139	-2.842	0.010
m16	as.factor(odd_even_factor)odd	0.313	0.200	1.563	0.133
m16	as.factor(vessel)Medeia	0.101	0.234	0.431	0.671
m16	as.factor(vessel)NW Explorer	-0.345	0.185	-1.863	0.077
m16	adj_raw_pink_log	0.309	0.082	3.774	0.001
m17	(Intercept)	6.787	1.021	6.647	0.000
m17	SEAK_SST_AMJ	-0.539	0.134	-4.030	0.001
m17	as.factor(odd_even_factor)odd	0.272	0.178	1.532	0.140
m17	as.factor(vessel)Medeia	0.154	0.203	0.761	0.455
m17	as.factor(vessel)NW Explorer	-0.292	0.165	-1.765	0.092
m17	adj_raw_pink_log	0.346	0.074	4.684	0.000
m18	(Intercept)	7.115	1.269	5.607	0.000
m18	SEAK_SST_AMJJ	-0.505	0.145	-3.490	0.002
m18	as.factor(odd_even_factor)odd	0.287	0.188	1.524	0.142
m18	as.factor(vessel)Medeia	0.152	0.219	0.695	0.495
m18	as.factor(vessel)NW Explorer	-0.344	0.173	-1.987	0.060
m18	adj_raw_pink_log	0.330	0.078	4.256	0.000

Table 7: Parameter estimates for the 18 individual models for Models m1a-m18a.

Model	Term	Estimate	Standard Error	Statistic	p value
m1a	(Intercept)	2.496	0.163	15.298	0.000
m1a	CPUE	0.304	0.072	4.201	0.000
m1a	as.factor(odd_even_factor)odd	0.400	0.175	2.289	0.030
m2a	(Intercept)	6.028	1.353	4.456	0.000
m2a	CPUE	0.396	0.074	5.343	0.000
m2a	as.factor(odd_even_factor)odd	0.196	0.176	1.112	0.277
m2a	ISTI20_JJ	-0.368	0.140	-2.627	0.015
m3a	(Intercept)	5.528	0.693	7.973	0.000
m3a	CPUE	0.363	0.057	6.400	0.000
m3a	as.factor(odd_even_factor)odd	0.357	0.134	2.670	0.013
m3a	Chatham_SST_May	-0.417	0.094	-4.445	0.000
m4a	(Intercept)	5.832	1.221	4.776	0.000
m4a	CPUE	0.334	0.066	5.099	0.000
m4a	as.factor(odd_even_factor)odd	0.346	0.157	2.199	0.037
m4a	Chatham_SST_MJJ	-0.345	0.125	-2.751	0.011
m5a	(Intercept)	6.125	0.852	7.185	0.000
m5a	CPUE	0.360	0.057	6.269	0.000
m5a	as.factor(odd_even_factor)odd	0.332	0.136	2.442	0.022
m5a	Chatham_SST_AMJ	-0.486	0.113	-4.304	0.000
m6a	(Intercept)	6.217	1.089	5.709	0.000
m6a	CPUE	0.345	0.062	5.577	0.000
m6a	as.factor(odd_even_factor)odd	0.335	0.148	2.260	0.033
m6a	Chatham_SST_AMJJ	-0.434	0.126	-3.445	0.002
m7a	(Intercept)	5.221	0.647	8.067	0.000
m7a	CPUE	0.370	0.058	6.373	0.000
m7a	as.factor(odd_even_factor)odd	0.384	0.135	2.837	0.009
m7a	Icy_Strait_SST_May	-0.399	0.093	-4.293	0.000
m8a	(Intercept)	5.561	1.105	5.032	0.000
m8a	CPUE	0.341	0.066	5.193	0.000
m8a	as.factor(odd_even_factor)odd	0.362	0.156	2.320	0.029
m8a	Icy_Strait_SST_MJJ	-0.313	0.112	-2.798	0.010
m9a	(Intercept)	5.874	0.822	7.144	0.000
m9a	CPUE	0.361	0.058	6.186	0.000
m9a	as.factor(odd_even_factor)odd	0.388	0.137	2.831	0.009
m9a	Icy_Strait_SST_AMJ	-0.471	0.113	-4.159	0.000
m10a	(Intercept)	5.876	1.012	5.806	0.000
m10a	CPUE	0.350	0.063	5.588	0.000
m10a	as.factor(odd_even_factor)odd	0.365	0.148	2.461	0.021
m10a	Icy_Strait_SST_AMJJ	-0.398	0.118	-3.371	0.002

Model	Term	Estimate	Standard Error	Statistic	p value
m11a	(Intercept)	5.265	0.648	8.126	0.000
m11a	CPUE	0.334	0.056	5.960	0.000
m11a	as.factor(odd_even_factor)odd	0.393	0.134	2.924	0.007
m11a	NSEAK_SST_May	-0.378	0.087	-4.357	0.000
m12a	(Intercept)	5.629	1.118	5.034	0.000
m12a	CPUE	0.311	0.064	4.836	0.000
m12a	as.factor(odd_even_factor)odd	0.382	0.155	2.459	0.021
m12a	NSEAK_SST_MJJ	-0.318	0.112	-2.826	0.009
m13a	(Intercept)	5.857	0.819	7.153	0.000
m13a	CPUE	0.328	0.057	5.752	0.000
m13a	as.factor(odd_even_factor)odd	0.384	0.137	2.801	0.010
m13a	NSEAK_SST_AMJ	-0.450	0.108	-4.156	0.000
m14a	(Intercept)	5.924	1.029	5.755	0.000
m14a	CPUE	0.318	0.061	5.188	0.000
m14a	as.factor(odd_even_factor)odd	0.380	0.148	2.566	0.017
m14a	NSEAK_SST_AMJJ	-0.397	0.118	-3.361	0.002
m15a	(Intercept)	5.266	0.701	7.512	0.000
m15a	CPUE	0.329	0.058	5.699	0.000
m15a	as.factor(odd_even_factor)odd	0.398	0.139	2.862	0.008
m15a	SEAK_SST_May	-0.349	0.087	-4.021	0.000
m16a	(Intercept)	5.524	1.117	4.948	0.000
m16a	CPUE	0.300	0.065	4.640	0.000
m16a	as.factor(odd_even_factor)odd	0.392	0.156	2.505	0.019
m16a	SEAK_SST_MJJ	-0.290	0.106	-2.736	0.011
m17a	(Intercept)	5.787	0.866	6.686	0.000
m17a	CPUE	0.322	0.059	5.489	0.000
m17a	as.factor(odd_even_factor)odd	0.389	0.141	2.755	0.011
m17a	SEAK_SST_AMJ	-0.408	0.106	-3.847	0.001
m18a	(Intercept)	5.801	1.044	5.558	0.000
m18a	CPUE	0.307	0.062	4.941	0.000
m18a	as.factor(odd_even_factor)odd	0.393	0.150	2.619	0.015
m18a	SEAK_SST_AMJJ	-0.358	0.112	-3.195	0.004

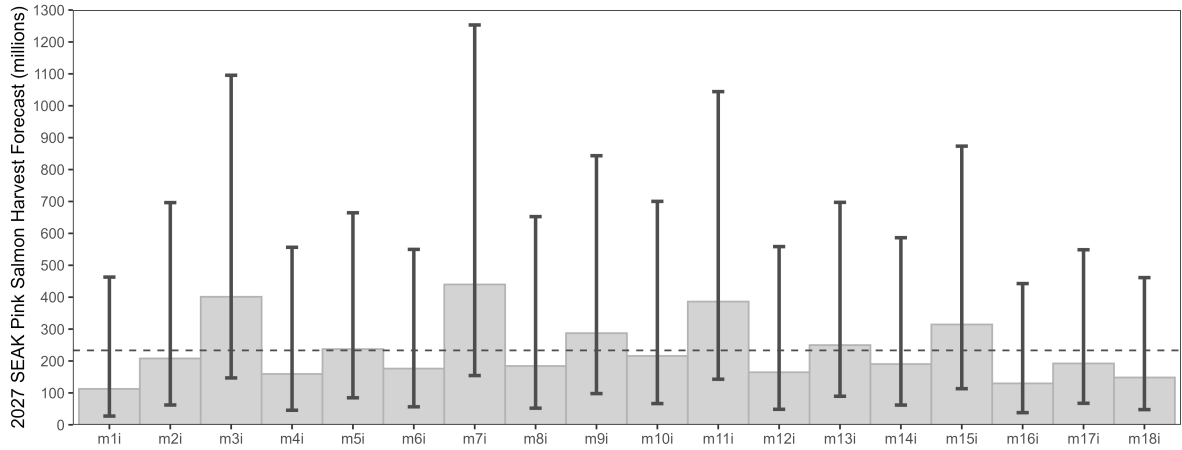


Figure B1: Bias-corrected forecasts (grey bars) for the 18 regression models (Models m1i-m18i) with 80% prediction intervals (vertical grey lines). A horizontal dotted line at the average model results is placed on the figure for reference only. The 2027-forecast using the model mxx has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).

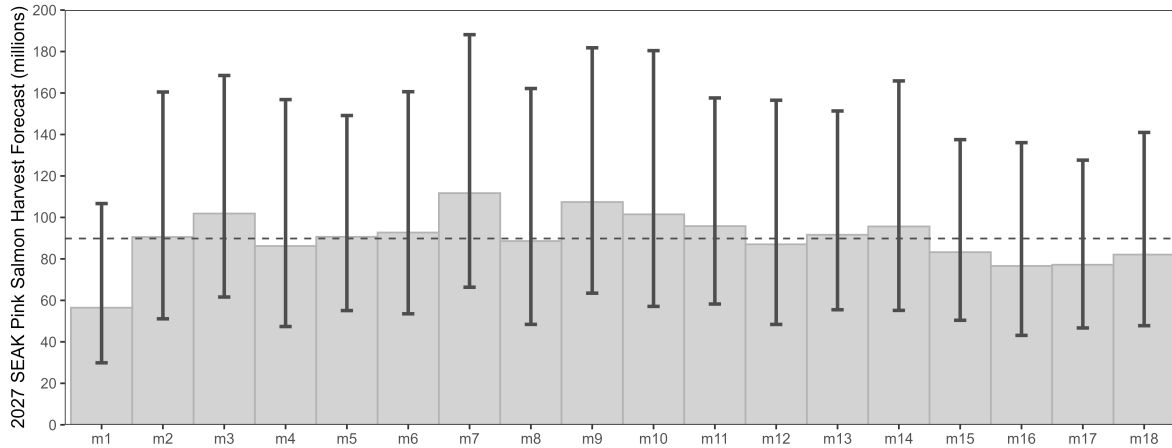


Figure B2: Bias-corrected forecasts (grey bars) for the 18 regression models (Models m1-m18) with 80% prediction intervals (vertical grey lines). A horizontal dotted line at the average model results is placed on the figure for reference only. The 2027-forecast using the model mxx has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).

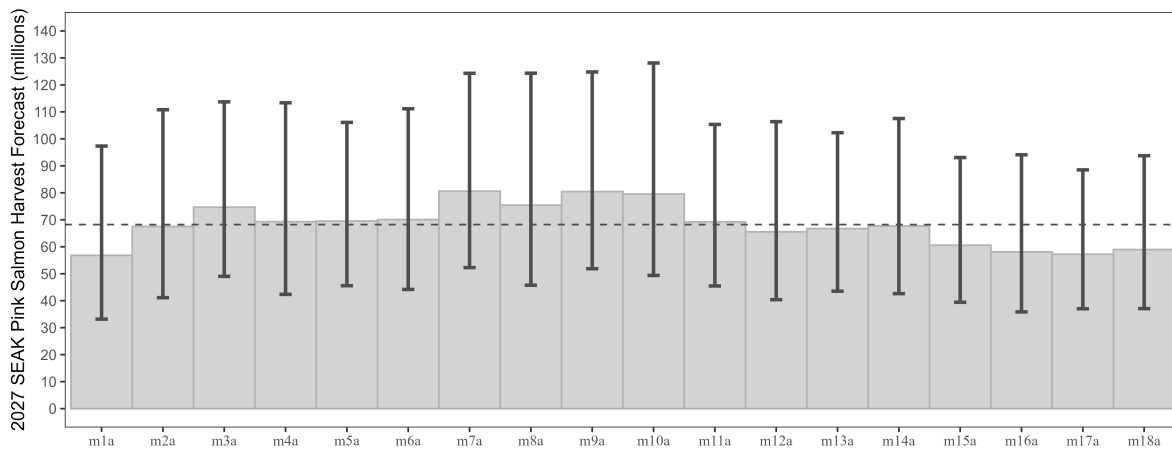


Figure B3: Bias-corrected forecasts (grey bars) for the 18 regression models (Models m1a-m18a) with 80% prediction intervals (vertical grey lines). A horizontal dotted line at the average model results is placed on the figure for reference only. The 2027-forecast using the model mxx has a point estimate of xx million fish (80% prediction interval: xx to xx million fish).